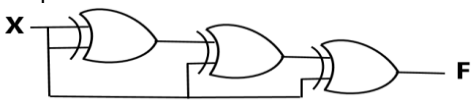




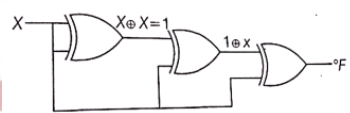
COMPUTER AWARENESS

- In IEEE single precision floating point representation, exponent is represented in.....
(a) 8 bit sign-magnitude representation
(b) 8 bit 2's complement representation
(c) Biases exponent representation with a bias value 127
(d) Biases exponent representation with a bias value 128
- With 4-bit 2's complement arithmetic, which of the following addition will result in overflow?
(a) 1111+1101 (b) 0110+0110
(c) 1101+0101 (d) 0101+1011
- If we can generate a maximum of 4 Boolean functions using n Boolean variables, what will be minimum value of n?
(a) 65536 (b) 16
(c) 1 (d) 4
- If the 2's complement representation of a number is $(011010)_2$, what is its equivalent hexadecimal representation?
(a) $(110)_{16}$ (b) $(1A)_{16}$
(c) $(16)_{16}$ (d) $(26)_{16}$
- For the circuit shown below, the complement of the output

(a) 0 (b) X (c) X' (d) 1
- If N is a 16-bit signed integer, then 2's complement representation of N is $(F87B)_{16}$. Then, 2's complement representation of $8*N$ is
(a) $(C3D8)_{16}$ (b) $(187B)_{16}$
(c) $(F878)_{16}$ (d) $(987B)_{16}$
- The base (or radix) of the number system such that the following equation holds $312/20=13.1$ is
(a) 3 (b) 4 (c) 5 (d) 6
- Which of the following represents $(D4)_{16}$?
(a) $(4E)_{16} - (5B)_{16}$ (b) $(14E)_{16} - (7A)_{16}$
(c) $(15C)_{16} - (6D)_{16}$ (d) $(1E4)_{16} - (A7)_{16}$
- How many Boolean expressions can be formed with 3 Boolean variables?
(a) 16 (b) 1024
(c) 32 (d) 256
- In an 8 bit representation of computer system the decimal number 47 has to be subtracted from 38 and the result in binary 2's complement is ...
(a) 11110111 (b) 10001001
(c) 11111001 (d) 11110001

ANSWER

1.	C	2.	B	3.	C	4.	B	5.	D
6.	A	7.	C	8.	B	9.	D	10.	a

SOLUTION

- (c)
In IEEE single precision floating point representation, exponent is represented in Biases exponent representation. The biased exponent is used for representation of negative exponents. The biased exponent (E) of single precision number can be obtained as
 $E = e + 127$
- (b)
The rules for detecting overflow in a 2's complement sum are:
(i) If the sum of two positive numbers, yields a negative result;
 $(+A) + (+B) = -C$
(ii) If the sum of two negative numbers, yields a positive results;
 $(-A) + (-B) = +C$
(iii) otherwise, the sum has not overflowed.
From the given options, only option (b) will produce the result in overflow.
- (c)
On n Boolean variables, there are 2^{2^n} different Boolean functions. If Boolean functions are 4, then
 $2^{2^n} = 4$
 $2^{2^n} = 2^2$
On comparing both sides, when base are equal
 $2^n = 2$
 $2^n = 2^1$
So, $n = 1$
- (b)
To convert the binary number into hexadecimal, making the group of 4 bits each from right to left. If the left most group has less than 4 bits, put in the necessary numbers of leading 0's on the left.
Now each group will be converted to decimal number
 $(011010)_2 \rightarrow (?)_{16}$
 $(011010)_2 \rightarrow (?)_{16}$
 $\begin{array}{r} 0001 \\ 1 \\ \hline 1010 \\ 10=A \end{array}$
So, $(011010)_2 = (1A)_{16}$
- (d)

The complement of $(F = 0)$ is $(F = 1)$
- (a)
 $N = (F87B)_{16}$ is 1111 1000 0111 1011
The most significant bit in the binary representation is 1.
Which implies that the number is negative.
To get the value of the number perform the 2's.
Complement of the number. We get N as -1925 and $8*N$ as -15400.
Since, $8*N$ is also negative, we need to find 2's
Complement of it (-15400)
Binary of 15400 = 0011 1100 0010 1000
2's complement = 1100 0011 1101 1000 = $(C3D8)_{16}$



7. (c)

Let x be the base of the number system.

The equation is

$$\frac{3x^2 + 1x^1 + 2x^0}{2x^1 + 0x^0} = 1x^1 + 3x^0 + 1x^{-1}$$

$$\frac{3x^2 + x + 2}{2x} = x + 3 + \frac{1}{x}$$

$$\frac{3x^2 + x + 2}{2x} = \frac{x^2 + 3x + 1}{x}$$

$$3x^2 + x + 2 = 2x^2 + 6x + 2$$

$$3x^2 - 2x^2 + x - 6x + 2 - 2 = 0$$

$$3x^2 - 2x^2 + x - 6x + 2 - 2 = 0$$

$$x^2 - 5x = 0$$

$$x(x - 5) = 0$$

$$x = 0 \text{ or } x = 5$$

As base of a number system cannot be zero.

So, $x = 5$.

8. (b)

First we convert all hexadecimal to decimal.

$$(D4)_{16} = D \times 16^1 + 4 \times 16^0$$

$$= 13 \times 16 + 4 \times 1$$

$$= 208 + 4 = (212)_{10}$$

This equation will be solved by option. So, we will take all option one by one.

Option (a)

$$(4E)_{16} = 4 \times 16^1 + E \times 16^0$$

$$= 4 \times 16 + 14 \times 1 = (78)_{10}$$

$$(5B)_{16} = 5 \times 16^1 + B \times 16^0$$

$$= 5 \times 16 + 11 \times 1 = (91)_{10}$$

$$(4E)_{16} - (5B)_{16} = (78)_{10} - (91)_{10}$$

$$= (-13)_{10}$$

This is wrong answer.

Option (b)

$$(14E)_{16} = 1 \times 16^2 + 4 \times 16^1 + E \times 16^0$$

$$= 1 \times 256 + 4 \times 16 + 14 \times 1$$

$$= (334)_{10}$$

$$(7A)_{16} = 7 \times 16^1 + A \times 16^0$$

$$= 7 \times 16 + 10 \times 1 = (122)_{10}$$

$$= (212)_{10}$$

This is right answer.

Option (c)

$$(15C)_{16} = 1 \times 16^2 + 5 \times 16^1 + C \times 16^0$$

$$= 1 \times 256 + 5 \times 16 + 12 \times 1 = (348)_{10}$$

$$(6D)_{10} = 6 \times 16^1 + 13 \times 1 = (109)_{10}$$

$$= (239)_{10}$$

This is wrong answer.

Option (d)

$$(1E4)_{16} = 1 \times 16^2 + E \times 16^1 + 4 \times 16^0$$

$$= 1 \times 256 + 14 \times 16 + 4 \times 1$$

$$= (484)_{10}$$

$$(A7)_{16} = A \times 16^1 + 7 \times 16^0$$

$$= 10 \times 16 + 7 \times 1 = (167)_{10}$$

$$(1E4)_{16} - (A7)_{16} = (484)_{10} - (167)_{10}$$

$$= (317)_{10}$$

This is wrong answer.

9. (d)

For the case of n Boolean variables, there are 2^{2^n}

Boolean expression can be formed.

If value of Boolean variable is 3.

Then Boolean expression will be 2^{2^n}

$$= 2^8 - 256$$

10. (a)

Binary representation of 47

$$= 101111$$

8 bit representation of 47

$$= 00101111$$

Binary representation of 38

$$= 100110$$

8 bit representation of 38

$$= 00100110$$

$$(47)_{10} - (38)_{10}$$

$$= 00101111$$

$$-00100110$$

$$00001001$$

$$1's \text{ complement} = 11110110$$

$$2's \text{ complement} = 11110110$$

$$+ \quad 1$$

$$11110111$$

